

MATH 3310 HOMEWORK ASSIGNMENT 5

DUE ON THURSDAY 29 SEPTEMBER 2022

Please turn in your homework in class or to Weeks Hall 347 no later than 5 pm.

- (1) Give two proofs of the logical equivalence

$$(p \implies q) \equiv (q \vee \sim p)$$

- (a) Using truth tables.
- (b) Using symbolic calculus in the style of Section 2.9 in the textbook.

- (2) State the negations of the following statements by reversing quantifiers:

- (a) For every rational number $q \neq 0$ the reciprocal $\frac{1}{q}$ is rational.
- (b) There exists a rational number q such that $q^2 = 5$.

- (3) Prove the following statements:

- (a) Let $n \in \mathbb{Z}$. If $|n - 1| \cdot |n + 1| < 0$, then $|n^2 - 1| > 1$.
- (b) Let $x \in \mathbb{R}$. If $x \in [0, 1]$, then $x^2 - 5x + 7 \neq 0$.

- (4) Prove the following statement:

Let x , y , and z be integers. If x and y are odd, then $xz + yz$ is even.

- (5) Prove the following statement:

$\forall x \in \mathbb{Z}$, $7x - 9$ is even if and only if x is odd.